William G. Underwood

Department of Statistics, University of Warwick, Coventry, CV4 7EZ, UK

`william.underwood@warwick.ac.uk` `wgunderwood.github.io`

Employment

Assistant Professor, Department of Statistics 2026–
University of Warwick

Postdoctoral Research Associate in Statistics 2024–2026
University of Cambridge

- Statistical Laboratory, DPMMS
- Advisor: Richard Samworth

Assistant in Instruction 2020–2024
Princeton University

Education

PhD in Operations Research & Financial Engineering 2019–2024
Princeton University

- Department of Operations Research & Financial Engineering
- Dissertation: Estimation and Inference in Modern Nonparametric Statistics
- Advisor: Matias Cattaneo

MA in Operations Research & Financial Engineering 2019–2021
Princeton University

MMath in Mathematics & Statistics 2015–2019
University of Oxford, St John's College

Research & publications

Articles

- Efficient and minimax optimal in-context nonparametric regression with transformers, with M. Ching, I. Popescu, N. Smith, T. Ma and R. J. Samworth. *International Conference on Machine Learning*, forthcoming, 2026. `arXiv:2601.15014`.
- Inference with Mondrian random forests, with M. D. Cattaneo and J. M. Klusowski. *Journal of the Royal Statistical Society, Series B 88*(3), 1060–1085, 2026. `arXiv:2310.09702`.
- Sharp anti-concentration inequalities for extremum statistics via copulas, with M. D. Cattaneo and R. P. Masini. *Bernoulli 32*(3), 2022–2045, 2026. `arXiv:2502.07699`.
- Yurinskii's coupling for martingales, with M. D. Cattaneo and R. P. Masini. *Annals of Statistics 53*(5), 2179–2203, 2025. `arXiv:2210.00362`.
- Uniform inference for kernel density estimators with dyadic data, with M. D. Cattaneo and Y. Feng. *Journal of the American Statistical Association 119*(548), 2695–2708, 2024. `arXiv:2201.05967`.
- Motif-based spectral clustering of weighted directed networks, with A. Elliott and M. Cucuringu. *Applied Network Science 5*(62), 2020. `arXiv:2004.01293`.
- Simple Poisson PCA: an algorithm for (sparse) feature extraction with simultaneous dimension determination, with L. Smallman and A. Artemiou. *Computational Statistics* 35, 559–577, 2019.

Preprints

- Upgrading survival models with CARE, with H. W. J. Reeve, O. Y. Feng, S. A. Lambert, B. Mukherjee and R. J. Samworth. *Preprint*, 2025. `arXiv:2506.23870`.

Working papers

- Higher-order Yurinskii coupling, with M. D. Cattaneo and R. P. Masini.
- Optimal polyadic estimation.

Presentations & conferences

- Statistical Sciences Research Institute Seminar, University of Southampton — Oct 2026
- Royal Statistical Society International Conference, Bournemouth — Sep 2026
- International Conference on Statistics and Data Science, Seville — Dec 2025
- GSEM Statistics Seminar, Université de Genève — Nov 2025
- MRC Biostatistics Unit Armitage Workshop, University of Cambridge — Oct 2025
- Economics Seminar, University of York — Oct 2025
- MRC Biostatistics Unit Seminar, University of Cambridge — Sep 2025
- StatMathAppli, Fréjus — Sep 2025
- London Symposium on Information Theory, Cambridge — May 2025
- International Conference on Statistics and Data Science, Nice — Dec 2024
- Statistics Seminar, University of Pittsburgh — Feb 2024
- Statistics Seminar, University of Illinois Urbana-Champaign — Jan 2024
- Statistics Seminar, University of Michigan — Jan 2024
- Eighth Princeton Day of Statistics, Princeton University — Nov 2023
- PhD Poster Session, Two Sigma Investments, New York — Jul 2023
- Statistical Foundations of Data Science, Princeton University — May 2023
- Research Symposium, Two Sigma Investments, New York — Jun 2022
- Seventh Princeton Day of Statistics, Princeton University — Oct 2021
- Statistics Laboratory, Princeton University — Sep 2021

Software

- care-survival: upgrading survival models with CARE in Python, 2025.
  GitHub: `wgunderwood/care-survival`
- tex-fmt: LaTeX formatter written in Rust, 2024. GitHub: `wgunderwood/tex-fmt`
- MondrianForests: Mondrian random forests in Julia, 2023.
  GitHub: `wgunderwood/MondrianForests.jl`
- DyadicKDE: dyadic kernel density estimation in Julia, 2022.
  GitHub: `wgunderwood/DyadicKDE.jl`
- motifcluster: motif-based spectral clustering in R, Python and Julia, 2020.
  GitHub: `wgunderwood/motifcluster`

Awards & funding

- Early Career Researcher Grant, G-Research — 2025
- School of Engineering and Applied Science Award for Excellence, Princeton University — 2022
- Francis Robbins Upton Fellowship in Engineering, Princeton University — 2019
- Royal Statistical Society Prize, Royal Statistical Society & University of Oxford — 2019
- Gibbs Statistics Prize, University of Oxford — 2019
- James Fund for Mathematics Research Grant, St John's College, University of Oxford — 2017
- Casberd Scholarship, St John's College, University of Oxford — 2016

Teaching

Lecturer
University of Warwick

- ST343: Topics in Data Science — Spring 2027
- ST925: Statistical Theory of Neural Networks — Spring 2027
- ST921: Statistical Frontiers — Autumn 2026

Lecturer
University of Cambridge

- Part III Statistical Learning in Practice — Lent 2026

Supervisor
University of Cambridge

- Part III Essay on Score Matching and Diffusion Models — Lent 2026
- Part II Probability & Measure — Michaelmas 2025
- Part IB Statistics — Lent 2025
- Part III Essay on Inference with Random Forests — Lent 2025

Drop-in Session Leader
University of Cambridge

- Part III Modern Statistical Methods — Lent 2025
- Part III Topics in Statistical Theory — Lent 2025
- Part III Concentration Inequalities — Lent 2025

Assistant in Instruction
Princeton University

- ORF 499: Senior Thesis — Spring 2024
- ORF 498: Senior Independent Research Foundations — Fall 2023
- SML 201: Introduction to Data Science — Fall 2023
- ORF 363: Computing and Optimization — Spring 2023
- ORF 524: Statistical Theory and Methods — Fall 2022
- ORF 526: Probability Theory — Fall 2022
- ORF 524: Statistical Theory and Methods — Fall 2021
- ORF 245: Fundamentals of Statistics — Spring 2021
- ORF 363: Computing and Optimization — Fall 2020

Professional experience

Quantitative Research Intern — 2023
Two Sigma Investments

Machine Learning Consultant — 2018
Mercury Digital Assets

Educational Consultant — 2018
Polaris & Dawn

Premium Tutor — 2016–2018
MyTutor

Statistics and Machine Learning Researcher — 2017–2017
Cardiff University

Data Science Intern — 2017
Rolls-Royce

Service

Peer review

Annals of the Institute of Statistical Mathematics; Annals of Statistics; Bernoulli; Biometrika; Econometric Theory; Information Theory, Probability and Statistical Learning; International Conference on Machine Learning; Journal of the American Statistical Association; Journal of Business & Economic Statistics; Journal of Causal Inference; Journal of Econometrics; Journal of Machine Learning Research; Journal of Nonparametric Statistics; Journal of the Royal Statistical Society, Series B; Journal of Statistical Computation and Simulation; Neural Information Processing Systems; Operations Research; SIAM Journal on Mathematics of Data Science; Statistical Science; Statistics and Computing; Technometrics.

Other service

- Organiser, Statistical Learning & Inference Seminar, University of Warwick 2026–2027
- Organiser, Statistics Department Conference, University of Warwick 2026–2027
- Consultant, Cambridge Statistics Clinic, University of Cambridge 2024–2026
- Local organiser, Statistical Foundations of Data Science, Princeton University 2023

References

- Richard Samworth, Professor, Statistical Laboratory, DPMMS, University of Cambridge
- Matias Cattaneo, Professor, Department of Economics, Princeton University
- Jianqing Fan, Professor, ORFE, Princeton University
- Jason Klusowski, Associate Professor, ORFE, Princeton University